

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

CO1 ASSESSMENT TOOL 1 – Case Study Analysis

Course Code:	CSA10 / CSA1063	Course Title:	Software Engineering
CO Assessed:	CO1	Assessment:	Assessment Tool 1-Case Study Analysis
Weightage:	20%	Total Marks:	25
CO2 Statement:	<i>Perform detailed requirements analysis, design scalable architectures, and prioritize features with a focus on cross-disciplinary skills</i>		
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Date:	22-07-26	Duration:	60 minutes

Code of Conduct

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____

AN AI-DRIVEN HOTEL RESERVATION AND ROOM MANAGEMENT PLATFORM WITH DYNAMIC PRICING AND DEMAND FORECASTING

Question

Based on the **assigned problem statement**, perform a **Scenario-Based Requirement Analysis** and prepare a Software Requirement Specification (SRS) document. Your analysis should include the following:

1. Understand the Problem Statement

○ Explain the Problem in Your Own Words

Small hotels often use fixed room prices throughout the year without considering changes in customer demand.

- This pricing method reduces their ability to earn maximum profit during weekends, holidays, and festival seasons when demand is high.
- During off-season periods, hotels may experience low occupancy because room prices are not reduced to attract more customers. Most small hotels also rely on manual booking systems such as registers or spreadsheets.
- Manual booking and pricing require extra time and effort from hotel staff. These manual processes increase the chances of errors such as incorrect pricing, double bookings, and delayed room updates.
- Customers may also face inconvenience because they cannot easily check room availability or make online reservations. The proposed system aims to automate hotel room booking and room management using a software platform. It introduces dynamic pricing, where room rates automatically change based on demand, occupancy, and booking dates. The system also allows customers to search, book, modify, and cancel reservations online with ease. Hotel administrators can manage rooms, bookings, pricing, and customer details

through a centralized dashboard. This reduces manual work and improves operational efficiency.

- **Identify the Objectives of the Proposed System**

- To automate the hotel room booking process.
- To reduce manual work in managing bookings and room prices.
- To allow customers to book and cancel rooms online easily.
- To automatically adjust room prices based on demand and occupancy.
- To provide real-time room availability information.
- To improve hotel revenue and customer satisfaction.
- To provide a secure, reliable, and user-friendly hotel management system.

- **State the Scope of the Problem**

The **SmartStay Hotel Reservation and Room Management System** is designed mainly for small and medium-sized hotels that do not have an automated room booking and pricing system. The project focuses on providing customers with an online platform to search for available rooms, make reservations, modify or cancel bookings, and complete secure online payments. It also includes an administrator dashboard that enables hotel staff to manage room details, bookings, customer information, and pricing efficiently. A key feature of the system is **dynamic pricing**, which automatically adjusts room rates based on customer demand, occupancy levels, weekends, holidays, and booking dates. The system aims to reduce manual work, minimize booking errors, improve room occupancy, and increase hotel revenue. However, the project is limited to hotel reservation and room management for individual small hotels and does not include features such as restaurant management, airline booking, or large multi-branch hotel chain management. Future versions of the system can be enhanced with AI-based demand forecasting, mobile applications, and support for multiple hotel branches.

2. Identify Key Stakeholders and Challenges

○ Identify All the Stakeholders Involved in the System

The SmartStay system involves several stakeholders who play different roles in its operation. The primary stakeholders are customers, hotel administrators, and hotel staff, as they directly interact with the system. Customers use the system to search for rooms, make reservations, cancel bookings, and complete online payments. Hotel administrators manage room details, bookings, pricing, and customer information through the admin dashboard. Hotel staff handle check-in, check-out, and room status updates. Secondary stakeholders include the payment gateway, which securely processes online transactions, the developer who develops and maintains the software, and the hotel owner who monitors the business performance and benefits from increased revenue and improved hotel operations. The SmartStay system shall allow customers to register and create an account. The system shall provide secure login and logout functionality for users and administrators. Customers shall be able to search available rooms based on dates and room type. The system shall display room details, prices, and availability in real time. Customers shall be able to book hotel rooms online. Users shall have the option to modify or cancel their reservations before check-in. The system shall automatically update room availability after every booking or cancellation. Dynamic pricing shall automatically adjust room rates based on demand, occupancy, and booking date. The system shall generate booking confirmations and payment receipts. Customers shall be able to make secure online payments using different payment methods. Hotel administrators shall manage rooms, bookings, pricing, and customer records through an admin dashboard. Staff shall update room status during check-in and check-out. The system shall generate reports on occupancy, bookings, and revenue.

○ Explain the Roles and Responsibilities of Each Stakeholder

1. Customer

- Registers and logs into the system.
- Searches for available rooms.
- Books, modifies, or cancels reservations.

- Makes secure online payments.
- Views booking history and receives booking confirmations.

2. Hotel Administrator

- Manages room details and availability.
- Updates room prices and dynamic pricing settings.
- Monitors and manages customer bookings.
- Maintains customer records and hotel information.
- Generates occupancy, booking, and revenue reports.

3. Hotel Staff

- Performs guest check-in and check-out.
- Updates room status after cleaning or maintenance.
- Assists customers during their stay.
- Verifies reservation details.
- Reports room availability to the administrator.

4. Payment Gateway

- Processes online payments securely.
- Verifies payment transactions.
- Confirms successful or failed payments.
- Supports multiple payment methods.
- Ensures secure financial transactions.

5. Developer

- Designs and develops the SmartStay system.
- Tests the software for errors and bugs.
- Maintains and updates the application.
- Improves system performance and security.
- Adds new features and enhancements when required.

6. Hotel Owner

- Monitors overall hotel performance.
- Reviews occupancy and revenue reports.
- Makes business decisions based on system data.
- Ensures smooth operation of hotel services.

- Benefits from increased revenue and improved customer satisfaction.
- **Discuss the Major Challenges and Issues Faced by the Stakeholders**

Different stakeholders face different challenges while using the hotel reservation system. Customers may experience fluctuating room prices during peak seasons and payment failures during online transactions. Hotel administrators face difficulties in managing room availability and pricing if updates are not performed correctly. Hotel staff may encounter problems when handling large numbers of bookings during busy periods.

- Payment gateway services may experience transaction delays or technical failures. Developers must ensure that the dynamic pricing algorithm works accurately while maintaining system performance and security.
- Hotel owners may lose revenue if pricing decisions are not optimized or if room occupancy remains low during off-season periods. The payment gateway securely processes online payments and confirms successful transactions.
- Developers are responsible for designing, developing, testing, maintaining, and updating the software. Hotel owners monitor occupancy, revenue, customer satisfaction, and overall hotel performance using reports generated by the system.

3. Analyze the Current Approach

- **Describe the Existing System or Current Process**

Most small hotels currently use manual methods for managing room bookings and pricing. Booking information is maintained using paper registers or spreadsheets, and room availability is updated manually by hotel staff. Room prices generally remain fixed throughout the year without considering customer demand or occupancy. Staff members manually confirm bookings and update room status after each reservation. This process consumes time, increases workload, and often leads to errors such as double bookings and incorrect room allocation. The existing system lacks automation, real-time updates, and intelligent pricing, making hotel management less

efficient.. The AI Pricing Engine automatically calculates room prices based on occupancy levels, customer demand, weekends, holidays, and seasonal trends. The Email/SMS Notification Service sends booking confirmations, payment receipts, booking reminders, cancellation notifications, and promotional offers to customers. The Database Server stores customer information, room details, booking records, payment history, pricing data, and hotel reports securely. Developers also interact with the system during software maintenance, testing, debugging, and future upgrades. Each actor performs a specific role that contributes to the successful functioning of the SmartStay platform. Together, these actors ensure efficient hotel reservation, secure transactions, effective room management, and improved customer satisfaction.

○ **Identify Its Strengths and Limitations**

The current system has a few advantages because it is simple to operate and requires very little technical knowledge. Small hotels can manage bookings manually without investing in expensive software. Hotel staff also have complete control over room allocation and pricing decisions. However, the system has many limitations, including fixed pricing, manual booking management, slow updates, and a high risk of human errors.

- It cannot automatically adjust room prices according to demand, resulting in lower profits during peak seasons and vacant rooms during off-season periods. These limitations reduce operational efficiency and customer satisfaction. Each actor in the SmartStay system performs specific use cases that define their interaction with the software. The Customer can create a new account by registering with personal details and logging into the system securely.
- After logging in, the customer searches for available rooms by selecting check-in and check-out dates, room type, occupancy, and preferred price range. The customer can view room details, compare prices, make online reservations, modify booking details, or cancel reservations according to hotel policies.
- Customers can complete secure online payments using debit cards, credit cards, UPI, or internet banking and receive booking confirmations through email or SMS. They can also view their booking history, payment records, and personal profile information.

- **Analyze the Problems or Gaps That Need to Be Addressed**

The major gap in the existing system is the lack of integration between booking management and room pricing. Since room prices remain fixed, hotels cannot maximize revenue during high-demand periods or attract customers during low-demand seasons. Manual booking increases the possibility of double bookings, incorrect room allocation, and delayed updates. Customers also face inconvenience because they cannot easily check room availability or complete online reservations. The absence of automation increases staff workload and reduces operational efficiency. Therefore, a modern software solution is required to automate hotel booking, room management, and dynamic pricing.

3. Propose a Suitable Solution Using Software Engineering Principles

- **Design a Software-Based Solution for the Given Problem**

- The proposed solution is SmartStay, an intelligent Hotel Reservation and Room Management System with Dynamic Pricing.
- The system allows customers to search for available rooms, book rooms, modify or cancel reservations, and make secure online payments through a web-based platform. It automatically adjusts room prices based on customer demand, occupancy, weekends, holidays, and booking dates.
- Hotel administrators can manage room details, bookings, pricing, and customer information through a centralized admin dashboard. The software reduces manual work, minimizes booking errors, improves room occupancy, and increases hotel revenue by automating the entire reservation process.

- **Describe the Major Functional Modules of the Proposed System**

1. User Module

- Allows customers to register and log in securely.
- Manages customer profiles and booking history.
- Enables users to modify personal information.

2. Room Management Module

- Maintains room categories and room details.

- Tracks room availability in real time.
- Updates room status after check-in and check-out.

3. Reservation Module

- Allows customers to search available rooms.
- Handles room booking, modification, and cancellation.
- Generates booking confirmation for customers.

4. Dynamic Pricing Module

- Automatically adjusts room prices based on demand.
- Considers occupancy, weekends, holidays, and booking dates.
- Helps maximize hotel revenue.

5. Payment Module

- Processes secure online payments.
- Supports multiple payment methods.
- Generates payment receipts after successful transactions.

6. Admin Dashboard

- Manages rooms, bookings, and customer information.
- Updates pricing policies.
- Generates occupancy and revenue reports.

○ Identify the Functional and Non-Functional Requirements

Functional Requirements

- User registration and secure login.
- Room search and availability checking.
- Online room booking and cancellation.
- Booking modification.
- Dynamic pricing based on demand.
- Secure online payment processing.
- Admin dashboard for managing rooms, bookings, and reports.

Non-Functional Requirements

- The system should provide fast response time.
- Customer data and payments should be secure.

- The software should be reliable with minimum downtime.
- The interface should be easy to use.
- The system should support future expansion.
- The software should be easy to maintain and update.
- The application should be available 24/7.

○ **Explain How Software Engineering Principles Are Incorporated into Your Solution**

Modularity

- The system is divided into independent modules such as booking, pricing, payment, and administration.
- Each module performs a specific function and can be maintained separately.

Scalability

- The system can support more customers as hotel demand increases.
- New features and additional hotel branches can be added easily.

Maintainability

- Clean and modular code allows developers to update the system without affecting other modules.
- Software maintenance becomes easier and faster.

Reliability

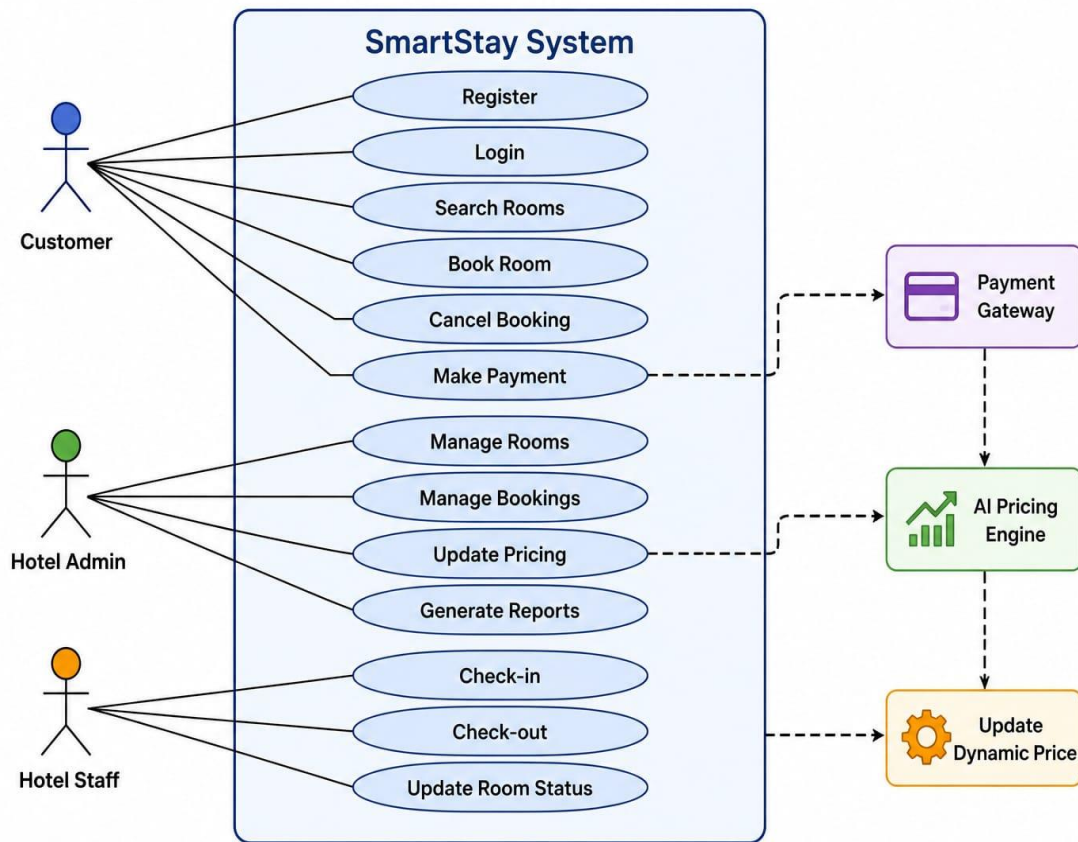
- The system provides accurate booking information and real-time room availability.
- Regular testing ensures smooth and error-free operation.

Usability

- The interface is simple, attractive, and easy for both customers and hotel staff.
- Users can complete bookings with minimal effort.

Security

- Customer passwords are encrypted.
- Secure authentication prevents unauthorized access.



4. Justify the Chosen Methodology/Framework

○ Select an Appropriate Software Development Life Cycle (SDLC) Mode

- Each module, such as room booking, dynamic pricing, payment processing, and admin management, can be developed, tested, and improved independently.
- Agile encourages continuous testing and regular feedback, helping developers identify and fix errors early in the development process.
- This approach makes the software more flexible, efficient, and suitable for projects where requirements may change during development.

- **Justify Why the Selected Methodology is Suitable**

- Agile is suitable for the SmartStay system because the project consists of multiple independent modules that can be developed separately. It allows developers to deliver working features quickly and improve them based on user feedback.
- Continuous testing in every sprint helps reduce software bugs and improves system quality. Agile also makes it easier to add new features, such as AI-based demand forecasting or mobile application support, without redesigning the entire system.
- The methodology improves collaboration among developers, testers, hotel administrators, and stakeholders, ensuring that the final software meets user requirements effectively.
- SmartStay is an AI-driven hotel reservation and room management system that automates hotel bookings, dynamic pricing, and reservation management. The system improves operational efficiency while providing customers with an easy online booking experience. The system also provides an administrator dashboard for managing rooms, customers, bookings, and hotel revenue.

- **Explain How It Supports Successful Project Development and Maintenance**

- Regular stakeholder feedback helps improve the software according to user needs and business requirements.
- Since the SmartStay system follows a modular structure, future maintenance becomes easier because updates can be made to individual modules without affecting the complete system.
- Agile also supports continuous improvement, making the software reliable, scalable, easy to maintain, and capable of adapting to future technological changes.

5. Justify the Chosen Methodology/Framework

- **Select an Appropriate Software Development Life Cycle (SDLC) Model**

- The Agile Software Development Model is selected for developing the SmartStay Hotel Reservation and Room Management System.

- Agile divides the project into small and manageable modules called sprints, allowing developers to complete one feature at a time. Each module, such as room booking, dynamic pricing, payment processing, and admin management, can be developed, tested, and improved independently.
 - Agile encourages continuous testing and regular feedback, helping developers identify and fix errors early in the development process. This approach makes the software more flexible, efficient, and suitable for projects where requirements may change during development.
- **Justify Why the Selected Methodology is Suitable**
- Agile is suitable for the SmartStay system because the project consists of multiple independent modules that can be developed separately.
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- **Explain How It Supports Successful Project Development and Maintenance**
- Agile supports successful project development by dividing the project into smaller tasks that are easier to design, develop, test, and implement. Frequent testing ensures that errors are identified and corrected before they affect other modules.
 - Regular stakeholder feedback helps improve the software according to user needs and business requirements. Since the SmartStay system follows a modular structure, future maintenance becomes easier because updates can be made to individual modules without affecting the complete system.
 - Agile also supports continuous improvement, making the software reliable, scalable, easy to maintain, and capable of adapting to future technological changes.

6. Discuss Expected Outcomes and Limitations

○ Describe the Expected Benefits of the Proposed Solution

- The SmartStay system is expected to improve hotel reservation and room management by automating manual processes. Customers can easily search for rooms, make reservations, cancel bookings, and complete secure online payments from anywhere.
- Dynamic pricing helps hotels maximize revenue during high-demand periods and improve room occupancy during low-demand seasons. The admin dashboard simplifies room management, booking management, and customer record maintenance. The system reduces booking errors, improves operational efficiency, saves time, and enhances overall customer satisfaction.

○ Explain How the Solution Addresses the Identified Challenges

- The SmartStay system solves the major problems of manual booking and fixed room pricing by integrating booking management with dynamic pricing.
- Room prices automatically increase during periods of high demand and decrease during low-demand periods, helping hotels maximize profits and reduce vacant rooms. Automated booking management updates room availability in real time, reducing errors such as double booking and incorrect room allocation.
- The admin dashboard simplifies hotel management by providing centralized control over rooms, bookings, pricing, and reports. Overall, the system improves efficiency, reduces manual work, and provides better services for both customers and hotel staff.

○ Discuss Potential Risks, Implementation Challenges, and Limitations

- The implementation of SmartStay may face challenges such as integrating secure payment gateways and maintaining accurate real-time room availability.
- System performance may be affected by internet connectivity issues or server failures. The dynamic pricing module currently follows predefined rules and may

not predict customer demand as accurately as advanced AI or machine learning models.

- The software is primarily designed for single hotels and does not fully support large hotel chains or multi-branch management. Regular maintenance, security updates, and staff training are also required to ensure smooth system operation and protect customer data.

- **Suggest possible future enhancements.**

AI-based demand forecasting using historical booking data for more accurate pricing, support for multiple hotel branches, integration with travel platforms like MakeMyTrip or Booking.com

Rubrics for Calculation

Criteria	Excellent (4 Marks)	Good (3 Marks)	Satisfactory (2 Marks)	Needs Improvement (1 Mark)
Scenario Understanding and Problem Analysis	Demonstrates thorough understanding of the scenario, stakeholders, objectives, and business context with clear analysis.	Demonstrates good understanding with minor omissions.	Basic understanding with limited analysis.	Poor understanding; major aspects missing or incorrect.
Functional and Non-Functional Requirement Identification	Clearly identifies comprehensive, accurate, and well-classified functional and non-functional requirements.	Identifies most requirements with minor classification errors.	Identifies only basic requirements; several omissions.	Requirements are incomplete, inaccurate, or poorly classified.

Actor Identification and Use Case Modeling	Correctly identifies all relevant actors and use cases; use case diagram is complete, accurate, and follows UML standards.	Most actors and use cases identified; diagram has minor errors.	Limited actors/use cases identified; diagram partially correct.	Actors/use cases missing or incorrect; diagram absent or inaccurate.
Software Requirement Specification (SRS) Documentation	SRS is well-structured, complete, professionally organized, and includes all required sections.	SRS is organized with minor missing sections or formatting issues.	SRS includes only essential sections with limited detail.	SRS is incomplete, poorly organized, or lacks essential components.
Assumptions, Constraints, and Requirement Validation	Clearly identifies realistic assumptions and constraints; validates requirements for completeness, consistency, and ambiguity with appropriate justification.	Identifies most assumptions and constraints; validation is adequate.	Limited assumptions/constraints; validation is superficial.	Assumptions, constraints, and validation are missing or incorrect.

Criteria	Mark
Scenario Understanding and Problem Analysis	/5
Functional and Non-Functional Requirement Identification	/5
Actor Identification and Use Case Modeling	/5
Software Requirement Specification (SRS) Documentation	/5
Assumptions, Constraints, and Requirement Validation	/5
TOTAL	/5

